

protected from this form of detection by their host programs. All of the methods used to detect and prevent viruses are also effective against the other forms of attack, therefore, the remainder of this paper will deal with the more serious problem of viral attacks.

Unix Virus Attacks

The promotion of the concept of “magical immunity” to computer viral attacks surfaces on a regular basis. This concept, while desirable, is misleading and dangerous since it tends to mask a real threat. Opponents of the possibility of viral attacks in Unix state that hardware instructions and operating system concepts such as supervisor mode or permission settings, security ratings like C2 or B1 provide protection. These ideas have been proven wrong by real life. The use of supervisor mode, the additional levels of protection provided by C2 and the mandatory access control provided by security level B1 are not necessary for viral activity and are therefore moot as a method of protection. This fact is supported by the existence of viruses that infect Unix systems as both scripts and binary.

In fact, virus attacks against Unix systems will eventually become more popular as simpler forms of attack become obsolete. Computer viruses have significantly more virility, methods of protection and opportunity for infection. Methods of protection have been highly refined in viruses, including rapid reproduction by infection, migration though evaluation of its environment, (boot viruses look for uninfected floppy diskettes) armor, stealth and polymorphism. In addition, the host system itself becomes a method of protection and propagation. Virus infected files are protected just as much by the operating system as are non-infected files. Introduction of viruses into systems have also been refined using technology called ‘droppers’. A dropper is a Trojan horse that has a virus or viruses as a payload. Finally, extensive networking technology such as NFS (Network File System) allows viruses to migrate between systems without effort.